In this chapter, you have learnt about:

- ✔ use of sensors, microprocessors and actuators in automated systems
- ✔ the advantages and disadvantages of automated systems in a number of key areas
- ✔ what is meant by robotics
- ✔ what characterises a robot
- ✔ the role of robots in a number of areas
- ✔ the advantages and disadvantages of robots in these areas
- ✔ the concept of artificial learning (AI)
- ✔ the main characteristics of AI
- ✔ expert systems
- ✔ machine learning.

Key terms used throughout this chapter

automated system – a combination of software and hardware designed and programmed to work automatically without the need for any human intervention

distributed control system (DCS) – a powerful computer system programmed to monitor and control a complex process without the need for human interaction

adaptive cruise control – the use of sensors, actuators and microprocessors to ensure that a vehicle keeps a safe distance behind another vehicle

accelerometer – a sensor that measures acceleration and deceleration and that can detect, for example, the orientation of a device

robotics – the branch of (computer) science that encompasses the design, construction and operation of robots

robot – a mechanical device that can carry out tasks normally done by humans

autonomous – able to operate independently without any human input

controller – a microprocessor that is in control of a process

WebCrawler/search bot – a software robot that roams the internet scanning websites and categorising them; often used by search engines

chatbots – a pop-up robot on a website that appears to enter into a meaningful conversation with a web user

end-effector – an attachment to a robot arm that allows it to carry out a specific task, such as spray painting

LiDaR – a contraction of light detection and ranging; the use of lasers to build up a 3D image of the surroundings

drone – a flying robot that can be autonomous or operated using remote control; a drone can be used for reconnaissance or deliveries

phenotyping – the process of observing the physical characteristics of a plant to assess its health and growth

cognitive – relating to the mental processes of the human brain involved in acquiring knowledge and understanding through thought, experiences and input from the five senses

artificial intelligence (AI) – a collection of rules and data which gives a computer system the ability to reason, learn and adapt to external stimuli

expert system – a form of AI that has been developed to mimic a human's knowledge and expertise

explanation system – part of an expert system which informs the user of the reasoning behind its conclusions and recommendations

inference engine – a kind of search engine used in an expert system which examines the knowledge base for information that matches the queries

inference rules – rules used by the inference engine and in expert systems to draw conclusions using IF statements

knowledge base – a repository of facts which is a collection of objects and attributes

object – an item stored in the knowledge base

attribute – something that defines the objects stored in a knowledge base

rules base – a collection of inference rules used to draw conclusions

machine learning – a sub-set of AI in which algorithms are trained and learn from past experiences and examples

web scraping – a method of obtaining data from websites